



## CS 540 Module Seven Assignment Guidelines and Rubric

### Overview

Understanding the difference between exploratory and explanatory analysis is a core skill for working with data. Being able to identify the purpose of each type of analysis and apply it appropriately improves the clarity and impact of your work. Professionals use these skills to inform decisions, create meaningful visualizations, and communicate insights to different audiences.

Knowing when to explore data and when to explain findings strengthens your ability to design effective workflows, justify your recommendations, and collaborate across teams. This kind of clarity supports communication in roles that involve research, analysis, and strategy. Practicing this distinction builds your confidence in choosing the right tools and techniques as you move forward in your studies and professional work.

### Scenario

Imagine you've been asked to brief a cross-functional team on your approach to analyzing a new data set in your professional field. Before the team moves forward, they need to understand the different ways data can be examined and shared. Your task is to write a short report that will be shared with the team.

### Directions

In this assignment, you will write a short report that explains the distinctions between exploratory and explanatory analysis, clarifies when each should be used, and highlights the value each provides in the decision-making process. Choose one of the following professional contexts as the foundation for your response: business operations, public health, education, environmental policy, or technology and software development. If you would like to use a professional context that is not listed, please reach out to your instructor for approval before beginning your assignment.

Specifically, you must address the following rubric criteria:

1. Define both **exploratory and explanatory analysis** in your own words. Include the following:
  - A. Relevant examples from your chosen context
2. Compare the two approaches by highlighting multiple **key differences** in how and why they are used.
3. Identify where each type of analysis fits within the broader **data analysis process**.
4. Explain how each type of analysis contributes to **decision making** in your chosen context.
5. Describe how understanding these distinctions can improve your ability to analyze and present data effectively in your **professional work**.
6. Provide an **example of a visualization** that fits each type of analysis in your chosen context.

## What to Submit

Submit your report as a 2- to 4-page Microsoft Word document with double spacing, 12-point Times New Roman font, and one-inch margins.

## AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

### Module Seven Assignment Rubric

| Criteria                                    | Exceeds Expectations (100%)                                                                    | Meets Expectations (90%)                                                                                                            | Partially Meets Expectations (70%)                                       | Does Not Meet Expectations (0%) | Value |
|---------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------|-------|
| <b>Exploratory and Explanatory Analysis</b> | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner | Defines both exploratory and explanatory analysis in own words, including relevant examples for the chosen context                  | Shows progress toward meeting expectations, but with errors or omissions | Does not attempt criterion      | 20    |
| <b>Key Differences</b>                      | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner | Compares the two approaches by highlighting multiple key differences in how and why they are used                                   | Shows progress toward meeting expectations, but with errors or omissions | Does not attempt criterion      | 15    |
| <b>Data Analysis Process</b>                | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner | Identifies where each type of analysis fits within the broader data analysis process                                                | Shows progress toward meeting expectations, but with errors or omissions | Does not attempt criterion      | 15    |
| <b>Decision Making</b>                      | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner | Explains how each type of analysis contributes to decision-making in the chosen context                                             | Shows progress toward meeting expectations, but with errors or omissions | Does not attempt criterion      | 15    |
| <b>Professional Work</b>                    | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner | Describes how understanding the distinctions can improve one's ability to analyze and present data effectively in professional work | Shows progress toward meeting expectations, but with errors or omissions | Does not attempt criterion      | 15    |

| Criteria                     | Exceeds Expectations (100%)                                                                     | Meets Expectations (90%)                                                                     | Partially Meets Expectations (70%)                                                                                                          | Does Not Meet Expectations (0%)                                        | Value |
|------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------|
| <b>Visualization Example</b> | Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner  | Provides an example of a visualization that fits each type of analysis in the chosen context | Shows progress toward meeting expectations, but with errors or omissions                                                                    | Does not attempt criterion                                             | 15    |
| <b>Clear Communication</b>   | Exceeds expectations with an intentional use of language that promotes a thorough understanding | Consistently and effectively communicates in an organized way to a specific audience         | Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding | Shows no evidence of consistent, effective, or organized communication | 5     |
| <b>Total:</b>                |                                                                                                 |                                                                                              |                                                                                                                                             |                                                                        | 100%  |